

SCAN Foundations

PSYCH 511.001 Spring 2026

2026-01-12

Foundations of Social, Cognitive, and Affective Neuroscience (SCAN)

Instructor

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<https://gilmore-lab.github.io>

Where & When

467 Moore Thursdays, 1-4 PM

About the course

The first scientific psychologists were physiologists fascinated by the possibility of understanding the mind by studying the brain. In this course, we will explore the historical roots and contemporary challenges associated with the study of biological approaches to complex adaptive behavior. In doing so, we will read and examine critically primary source readings that discuss basic patterns and processes of brain structure and function. The goal is to provide students with a basic foundation of knowledge about the structures and functions of the nervous system that can provide the basis for future study.

This course is one of two required courses for the [Specialization in Cognitive and Affective Neuroscience \(SCAN\)](#).

Prerequisites

Undergraduate coursework in neuroscience or physiological psychology such as the equivalents of PSYCH 260 or BIO 469/470.

Week 01: Jan 12-16

Thu, Jan 15

Topic(s)

- Structure of the course
- Levels of analysis
- Causality in brain and behavior
- Does neuroscience need behavior? What does psychology need?

Read/Watch

- Required
 - Krakauer et al. (2017)
 - Parada and Rossi (2018)
 - Newcombe (2017)
- Optional
 - Siddiqi et al. (2022)
 - Churchland and Sejnowski (1988)
 - Favela (2020)
 - Ross and Bassett (2024)
 - National Geographic (2014)

Materials

- [Slides](#)

Assignments

- *Assigned:* [Exercise LVL](#) • [Levels/Terms](#)
 - *Due:* 2026-01-22
 - [Canvas dropbox](#)

Week 02: Jan 19-23

Thu, Jan 22

Topics

- Neuroanatomy

Read/Watch

- Required
 - Leicester Medical School Radiology (2021) (Through ~ 8:20)
 - [Neuroanatomy notes](#)

Materials

- [Slides](#)
- [Allen Brain Atlas](#).

Assignments

- *Due:* [Exercise LVL](#) • [Levels/Terms](#)
 - [Canvas dropbox](#)
- *Assigned:* [Exercise ANAT](#) • [Neuroanatomy](#)
 - *Due:* 2026-01-29
 - [Canvas dropbox](#)

Week 03: Jan 26-30

Thu, Jan 29

Topic(s)

- Action

Read/Watch

- Required
 - Shenoy, Sahani, and Churchland (2013)
 - Goulding, Bollu, and Büschges (2025)
- Recommended
 - *Hardwired for Song and Dance: The Neurobiology of Language and Movement* (2023)

Assignments

- *Due:* [Exercise ANAT](#) • [Neuroanatomy](#)
 - [Canvas dropbox](#)

Week 04: Feb 2-6

Thu, Feb 05

Topic(s)

- Perception

Read/Watch

- Required
 - Khalsa et al. (2018)
 - Shenoy, Sahani, and Churchland (2013)

Assignment(s)

- *Assigned:* [Final project proposal](#)
 - *Due:* 2026-03-05
 - [Canvas dropbox](#)
- *Assigned:* [Exercise PA](#) • [Perception & Action](#)
 - *Due:* 2026-02-12.
 - [Canvas dropbox](#)

Week 05: Feb 9-13

Thu, Feb 12

Topic(s)

- Learning & memory

Read/Watch

Assignment(s)

- *Due:* [Exercise PA](#) • [Perception & Action](#)
 - [Canvas dropbox](#)

Week 06: Feb 16-20

Thu, Feb 19

Topic(s)

- Emotion

Read/Watch

- Required
 - Malezieux, Klein, and Gogolla (2023)
 - Watabe-Uchida, Eshel, and Uchida (2017)
 - Siegel et al. (2018)
- Recommended
 - LeDoux (2000)

Assignment(s)

- *Assigned:* [Exercise EMO](#) • [Measuring emotion](#)
 - *Due:* 2026-02-26
 - [Canvas dropbox](#)

Week 07: Feb 23-27

Thu, Feb 26

Topic(s)

- Sleep & circadian rhythms

Read/Watch

- Required
 - Nir and Tononi (2010)
 - Rasch and Born (2013)

Assignment(s)

- *Due:* [Exercise EMO](#) • [Measuring emotion](#)
 - [Canvas dropbox](#)
- *Assigned:* [Exercise ZZ](#) • [Sleep & Circadian Rhythms](#)
 - *Due:* 2026-03-05
 - [Canvas dropbox](#)

Readings

Materials

Week 08: Mar 2-6

Thu, Mar 5

Topic(s)

- Evolution of the nervous system

Read/Watch

- Required
 - Charvet and Finlay (2012)
 - Hofman (2014)
 - Sachkova, Modepalli, and Kittelmann (2025)
- Optional
 - Castrillon et al. (2023)

Assignment(s)

- *Due:* [Final project proposal](#)
 - [Canvas dropbox](#)
- *Due:* [Exercise ZZ • Sleep & Circadian Rhythms](#)
 - [Canvas dropbox](#)
- *Assigned:* [Exercise EVO • Evolution](#)
 - *Due:* 2026-03-19.
 - [Canvas dropbox](#)

Spring Break: Mar 9-13

NO CLASS

Week 09: Mar 16-20

Thu, Mar 19

Topic(s)

- Development of the nervous system

Read/Watch

- Required
 - Cao, Huang, and He (2017)
 - Blumberg and Adolph (2023)
- Optional
 - Larsen et al. (2023)
 - Rakic (2009)

Assignment(s)

- *Due:* [Exercise EVO](#) • [Evolution](#)
 - [Canvas dropbox](#)
- *Assigned:* [Exercise DEV](#) • [Development](#)
 - *Due:* 2026-03-26.
 - [Canvas dropbox](#)

Week 10: Mar 23-27

Thu, Mar 26

Topic(s)

- Cellular neuroscience I
 - Anatomy
 - Physiology
 - * Resting potential

Read/Watch

- [Cellular neuroscience notes](#) | [PDF](#) |
- Optional
 - Won et al. (2025)
 - Zeng and Sanes (2017)
 - Oliveira et al. (2015)
 - Distéfano-Gagné et al. (2023)

Assignment(s)

- *Due:* [Exercise DEV](#) • Development
 - [Canvas dropbox](#)
- *Assigned:* [Exercise PHYS](#) • Neurophysiology I
 - *Due:* 2026-04-02
 - [Canvas dropbox](#)

Week 11: Mar 30 - Apr 3

Thu, Apr 02

Topics

- Cellular neuroscience II
 - Action potential
 - Synaptic transmission

Read/Watch

- [Cellular neuroscience notes](#)

Assignment(s)

- *Due:* [Exercise PHYS](#) • Neurophysiology I
 - [Canvas dropbox](#)
- *Assigned:* [Exercise AP](#) • Neurophysiology II
 - *Due:* 2026-04-09
 - [Canvas dropbox](#)

Week 12: Apr 6-10

Thu, Apr 09

Topic(s)

- Neurochemistry
 - Neurotransmitters
 - Hormones
- Neurocomputing

Read/Watch

- [Neurochemistry notes](#)
- Optional: Sarkar et al. (2016).

Assignment(s)

- *Due:* [Exercise AP](#) • [Neurophysiology II](#)
 - [Canvas dropbox](#)
- *Assigned:* [Exercise CHEM](#) • [Neurochemistry](#)
 - *Due:* 2026-04-16.
 - [Canvas dropbox](#)

Week 13: Apr 13-17

Thu, Apr 16

Topic(s)

- Disorder & disease I

Read/Watch

- Required
 - Howes, Bukala, and Beck (2023)
 - Volk et al. (2015)

Assignment(s)

- *Due:* [Exercise CHEM](#) • [Neurochemistry](#)
 - [Canvas dropbox](#)

Week 14: Apr 20-24

Thu, Apr 23

Topic(s)

- Disorder & disease II

Read/Watch

- Required
 - Moncrieff et al. (2022)
 - Namkung, Kim, and Sawa (2017)

Week 15: Apr 27 - May 1

Thu, Apr 30

Topics

- Frontiers in the biology of behavior
- Beethoven and the Cerebral Symphony

Finals: May 4-8

Thu, May 07

- [Final Project](#) write-up due
 - [Canvas dropbox](#)

References

- Blumberg, Mark S, and Karen E Adolph. 2023. "Protracted Development of Motor Cortex Constrains Rich Interpretations of Infant Cognition." *Trends in Cognitive Sciences* 27 (3): 233–45. <https://doi.org/10.1016/j.tics.2022.12.014>.
- Cao, Miao, Hao Huang, and Yong He. 2017. "Developmental Connectomics from Infancy Through Early Childhood." *Trends in Neuroscience* 40 (8): 494–506. <https://doi.org/10.1016/j.tins.2017.06.003>.
- Castrillon, Gabriel, Samira Epp, Antonia Bose, Laura Fraticelli, André Hechler, Roman Belenya, Andreas Ranft, et al. 2023. "An Energy Costly Architecture of Neuromodulators for Human Brain Evolution and Cognition." *Science Advances* 9 (50): eadi7632. <https://doi.org/10.1126/sciadv.adi7632>.
- Charvet, Christine J, and Barbara L Finlay. 2012. "Chapter 4 - Embracing Covariation in Brain Evolution: Large Brains, Extended Development, and Flexible Primate Social Systems." In *Progress in Brain Research*, edited by Michel A Hofman and Dean Falk, 195:71–87. Elsevier. <https://doi.org/10.1016/B978-0-444-53860-4.00004-0>.
- Churchland, P S, and T J Sejnowski. 1988. "Perspectives on Cognitive Neuroscience." *Science* 242 (4879): 741–45. <https://www.ncbi.nlm.nih.gov/pubmed/3055294>.
- Distéfano-Gagné, Félix, Sara Bitarafan, Steve Lacroix, and David Gosselin. 2023. "Roles and Regulation of Microglia Activity in Multiple Sclerosis: Insights from Animal Models." *Nature Reviews. Neuroscience* 24 (7): 397–415. <https://doi.org/10.1038/s41583-023-00709-6>.
- Favela, Luis H. 2020. "Cognitive Science as Complexity Science." *Wiley Interdisciplinary Reviews. Cognitive Science* 11 (4): e1525. <https://doi.org/10.1002/wcs.1525>.
- Goulding, Martyn, Tejapratap Bollu, and Ansgar Büschges. 2025. "Sensory Feedback and the Dynamic Control of Movement." *Annual Review of Neuroscience* 48 (July): 405–23. <https://doi.org/10.1146/annurev-neuro-112723-042229>.
- Hardwired for Song and Dance: The Neurobiology of Language and Movement*. 2023. YouTube. https://www.youtube.com/watch?v=5F5_epP6bEg.
- Hofman, Michel A. 2014. "Evolution of the Human Brain: When Bigger Is Better." *Frontiers in Neuroanatomy* 8 (March). <https://doi.org/10.3389/fnana.2014.00015>.
- Howes, Oliver D, Bernard R Bukala, and Katherine Beck. 2023. "Schizophrenia: From Neurochemistry to Circuits, Symptoms and Treatments." *Nature Reviews. Neurology*, December. <https://doi.org/10.1038/s41582-023-00904-0>.
- Khalsa, Sahib S, Ralph Adolphs, Oliver G Cameron, Hugo D Critchley, Paul W Davenport, Justin S Feinstein, Jamie D Feusner, et al. 2018. "Interoception and Mental Health: A Roadmap." *Biological Psychiatry: Cognitive Neuroscience and Neuroimaging* 3 (June): 501–13. <https://doi.org/10.1016/j.bpsc.2017.12.004>.
- Krakauer, John W, Asif A Ghazanfar, Alex Gomez-Marin, Malcolm A MacIver, and David Poeppel. 2017. "Neuroscience Needs Behavior: Correcting a Reductionist Bias." *Neuron* 93 (3): 480–90. <https://doi.org/10.1016/j.neuron.2016.12.041>.
- Larsen, Bart, Valerie J Sydnor, Arielle S Keller, B T Thomas Yeo, and Theodore D Satterthwaite. 2023. "A Critical Period Plasticity Framework for the Sensorimotor-Association Axis of

- Cortical Neurodevelopment.” *Trends in Neurosciences* 46 (10): 847–62. <https://doi.org/10.1016/j.tins.2023.07.007>.
- LeDoux, J. E. 2000. “Emotion Circuits in the Brain.” *Annual Review of Neuroscience* 23 (March): 155–84. <https://doi.org/10.1146/annurev.neuro.23.1.155>.
- Leicester Medical School Radiology. 2021. *Introduction to MRI of the Brain*. Youtube. <https://www.youtube.com/watch?v=Co7Qyu21QKU>.
- Malezieux, Meryl, Alexandra S Klein, and Nadine Gogolla. 2023. “Neural Circuits for Emotion.” *Annual Review of Neuroscience* 46 (July): 211–31. <https://doi.org/10.1146/annurev-neuro-111020-103314>.
- Moncrieff, Joanna, Ruth E Cooper, Tom Stockmann, Simone Amendola, Michael P Hengartner, and Mark A Horowitz. 2022. “The Serotonin Theory of Depression: A Systematic Umbrella Review of the Evidence.” *Molecular Psychiatry*, July. <https://doi.org/10.1038/s41380-022-01661-0>.
- Namkung, Ho, Sun-Hong Kim, and Akira Sawa. 2017. “The Insula: An Underestimated Brain Area in Clinical Neuroscience, Psychiatry, and Neurology.” *Trends in Neurosciences* 40 (4): 200–207. <https://doi.org/10.1016/j.tins.2017.02.002>.
- National Geographic. 2014. “Beautiful 3-D Brain Scans Show Every Synapse | National Geographic.” YouTube. <https://www.youtube.com/watch?v=nvXuq9jRWKE>.
- Newcombe, N. 2017. “If Neuroscience Needs Behavior, What Does Behavioral Science Need?” *APS Observer* 31 (December). <https://www.psychologicalscience.org/observer/if-neuroscience-needs-behavior-what-does-behavioral-science-need>.
- Nir, Yuval, and Giulio Tononi. 2010. “Dreaming and the Brain: From Phenomenology to Neurophysiology.” *Trends in Cognitive Sciences* 14 (February): 88–100. <https://doi.org/10.1016/j.tics.2009.12.001>.
- Oliveira, João Filipe, Vanessa Morais Sardinha, Sónia Guerra-Gomes, Alfonso Araque, and Nuno Sousa. 2015. “Do Stars Govern Our Actions? Astrocyte Involvement in Rodent Behavior.” *Trends in Neurosciences* 38 (9): 535–49. <https://doi.org/10.1016/j.tins.2015.07.006>.
- Parada, Francisco J, and Alejandra Rossi. 2018. “If Neuroscience Needs Behavior, What Does Psychology Need?” *Frontiers in Psychology* 9 (March): 433. <https://doi.org/10.3389/fpsyg.2018.00433>.
- Rakic, Pasko. 2009. “Evolution of the Neocortex: A Perspective from Developmental Biology.” *Nature Reviews. Neuroscience* 10 (10): 724–35. <https://doi.org/10.1038/nrn2719>.
- Rasch, Björn, and Jan Born. 2013. “About Sleep’s Role in Memory.” *Physiological Reviews* 93 (April): 681–766. <https://doi.org/10.1152/physrev.00032.2012>.
- Ross, Lauren N, and Dani S Bassett. 2024. “Causation in Neuroscience: Keeping Mechanism Meaningful.” *Nature Reviews. Neuroscience*, January. <https://doi.org/10.1038/s41583-023-00778-7>.
- Sachkova, Maria, Vengamanaidu Modepalli, and Maike Kittelmann. 2025. “The Deep Evolutionary Roots of the Nervous System.” *Annual Review of Neuroscience* 48 (July): 311–29. <https://doi.org/10.1146/annurev-neuro-112723-040945>.
- Sarkar, Amar, Soili M Lehto, Siobhán Harty, Timothy G Dinan, John F Cryan, and Philip W J Burnet. 2016. “Psychobiotics and the Manipulation of Bacteria–Gut–Brain Signals.” *Trends in Neurosciences* 39 (11): 763–81. <https://doi.org/10.1016/j.tins.2016.09.002>.

- Shenoy, Krishna V, Maneesh Sahani, and Mark M Churchland. 2013. "Cortical Control of Arm Movements: A Dynamical Systems Perspective." *Annual Review of Neuroscience* 36 (July): 337–59. <https://doi.org/10.1146/annurev-neuro-062111-150509>.
- Siddiqi, Shan H, Konrad P Kording, Josef Parvizi, and Michael D Fox. 2022. "Causal Mapping of Human Brain Function." *Nature Reviews Neuroscience* 23 (6): 361–75. <https://doi.org/10.1038/s41583-022-00583-8>.
- Siegel, Erika H, Molly K Sands, Wim Van den Noortgate, Paul Condon, Yale Chang, Jennifer Dy, Karen S Quigley, and Lisa Feldman Barrett. 2018. "Emotion Fingerprints or Emotion Populations? A Meta-Analytic Investigation of Autonomic Features of Emotion Categories." *Psychological Bulletin* 144 (4): 343–93. <https://doi.org/10.1037/bul0000128>.
- Volk, Lenora, Shu-Ling Chiu, Kamal Sharma, and Richard L Huganir. 2015. "Glutamate Synapses in Human Cognitive Disorders." *Annual Review of Neuroscience* 38 (July): 127–49. <https://doi.org/10.1146/annurev-neuro-071714-033821>.
- Watabe-Uchida, Mitsuko, Neir Eshel, and Naoshige Uchida. 2017. "Neural Circuitry of Reward Prediction Error." *Annual Review of Neuroscience* 40 (July): 373–94. <https://doi.org/10.1146/annurev-neuro-072116-031109>.
- Won, Woojin, Mridula Bhalla, Jae-Hun Lee, and C Justin Lee. 2025. "Astrocytes as Key Regulators of Neural Signaling in Health and Disease." *Annual Review of Neuroscience* 48 (July): 251–76. <https://doi.org/10.1146/annurev-neuro-112723-035356>.
- Zeng, Hongkui, and Joshua R Sanes. 2017. "Neuronal Cell-Type Classification: Challenges, Opportunities and the Path Forward." *Nature Reviews. Neuroscience*, August. <https://doi.org/10.1038/nrn.2017.85>.